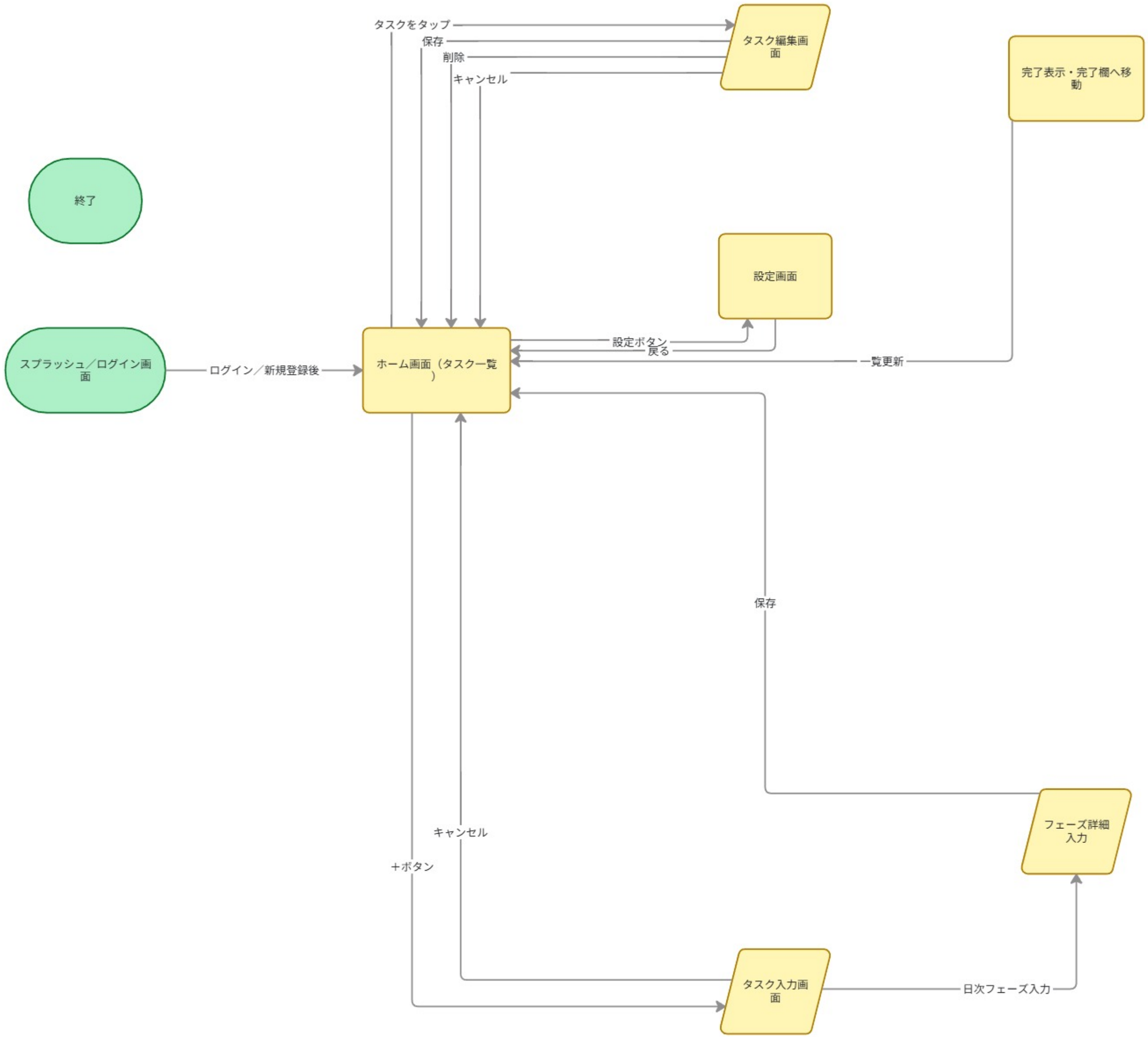
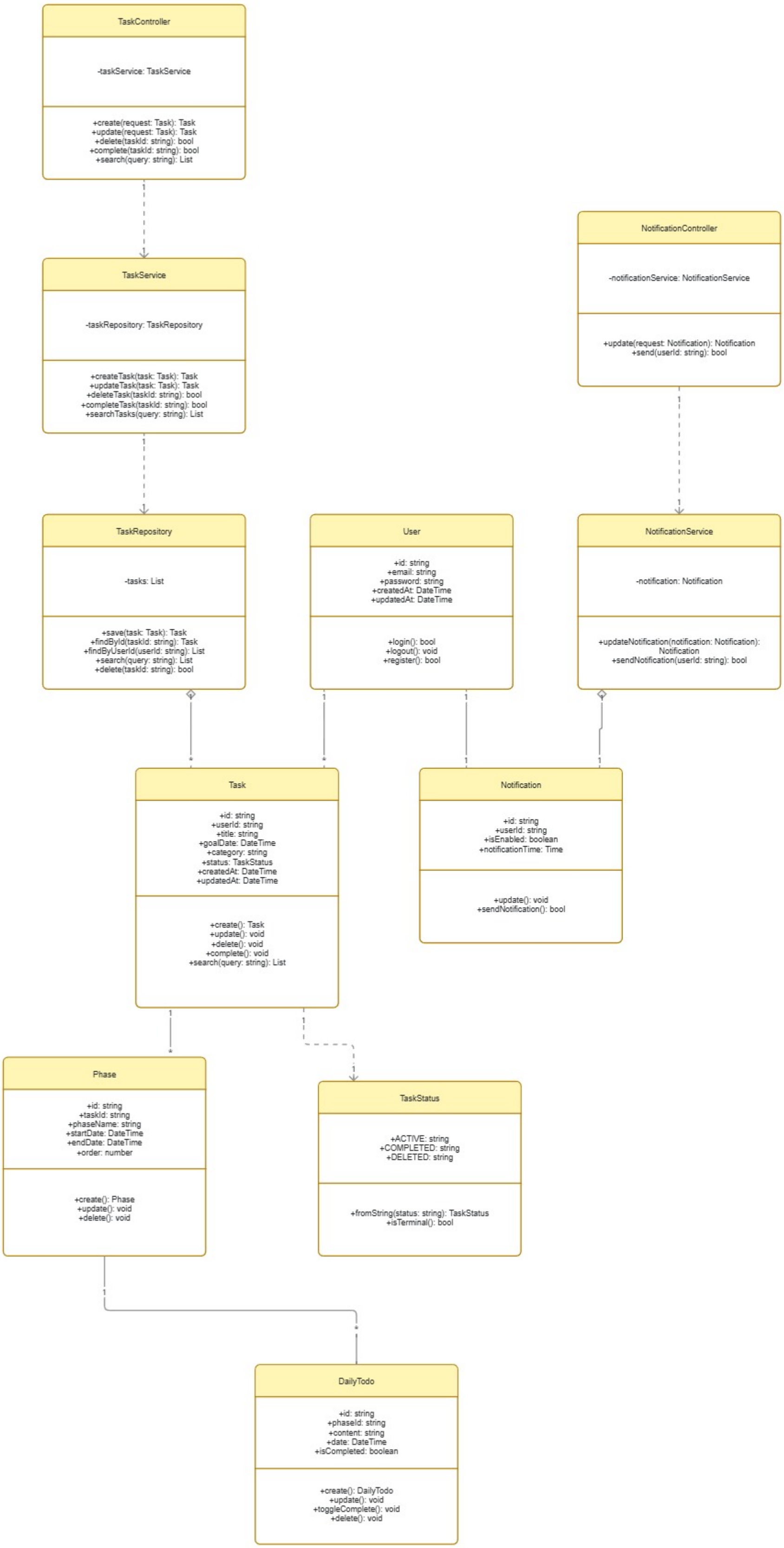
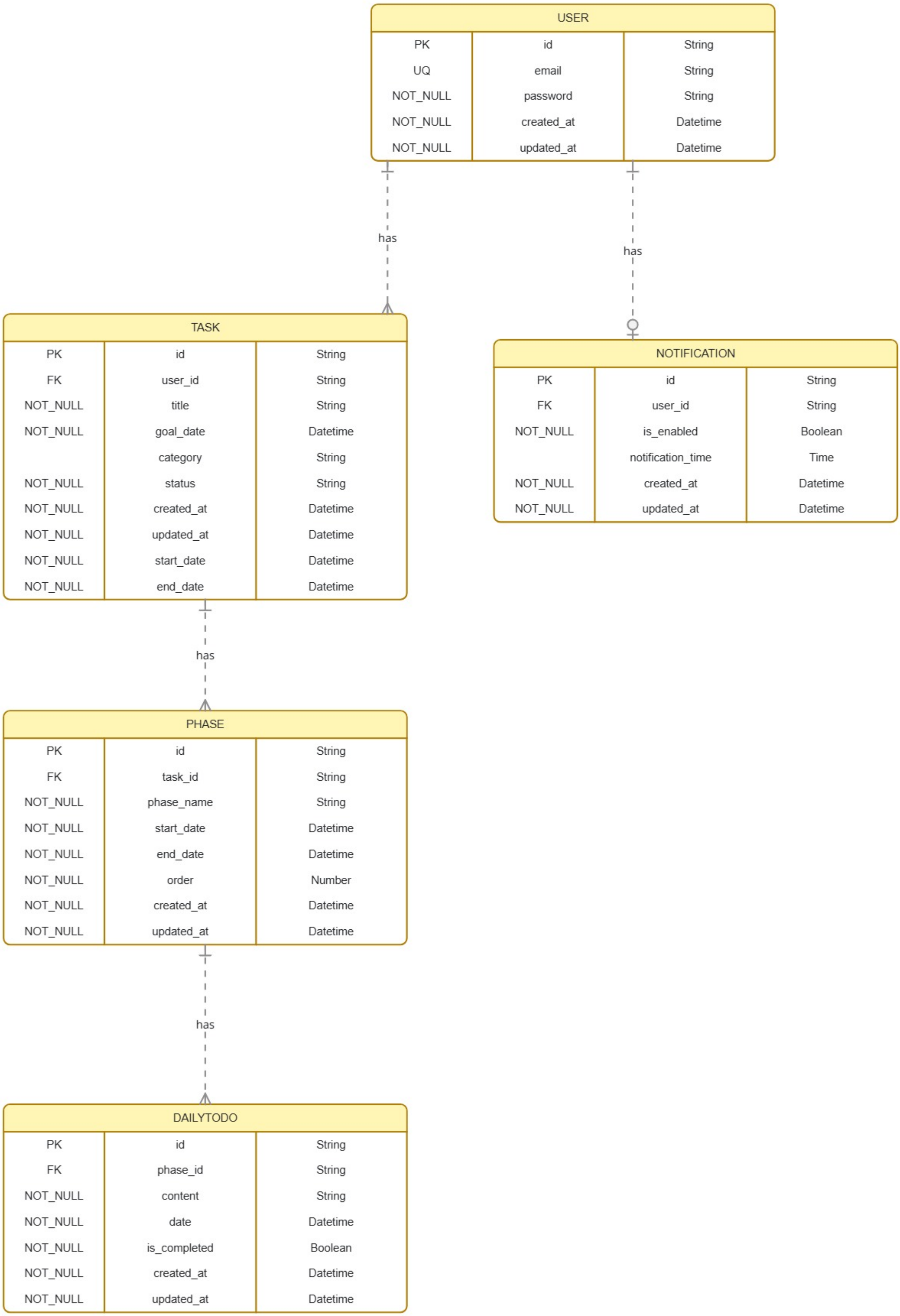
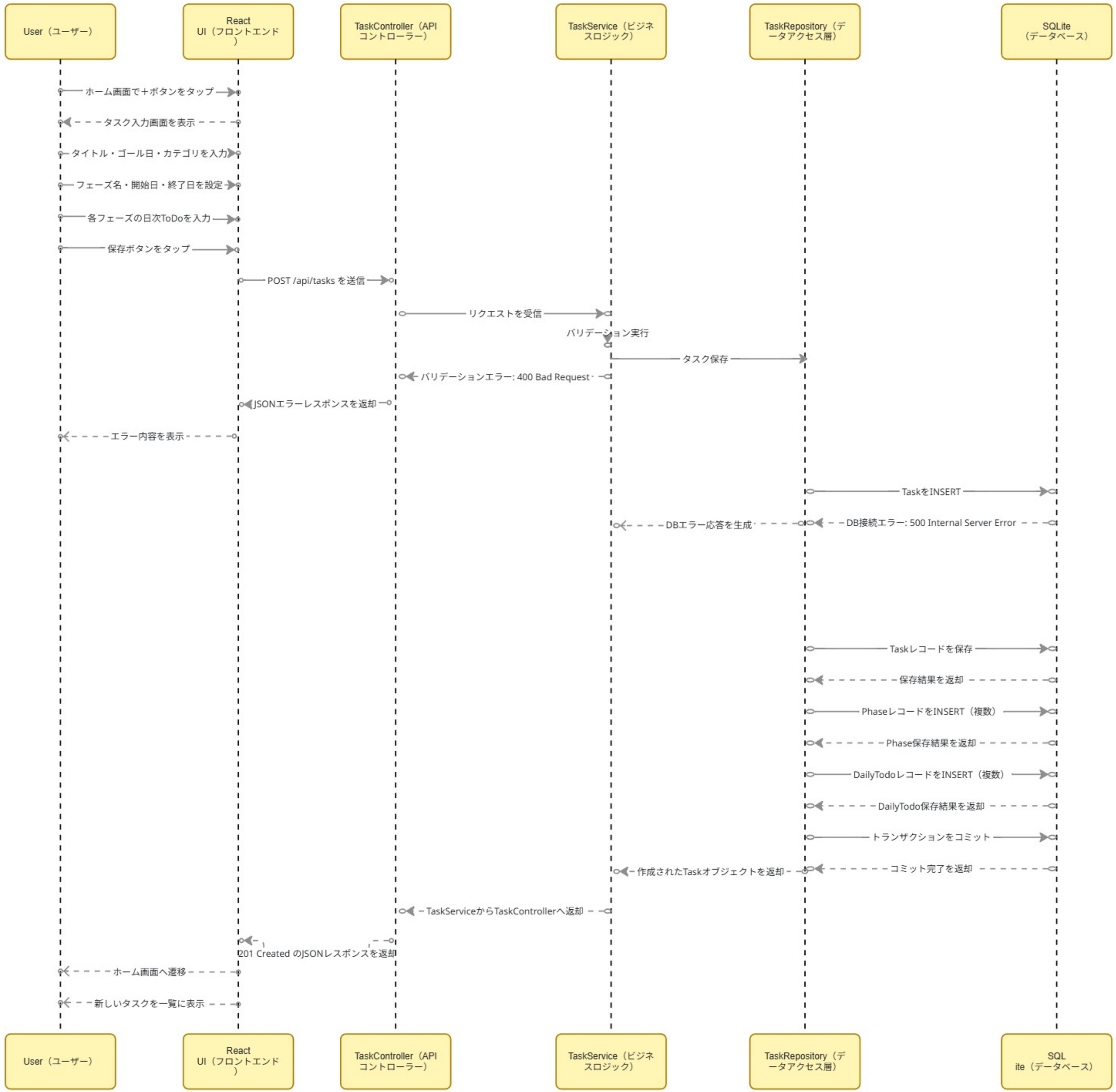

タスクをタップ
保存
削除
キャンセル
タスク編集画面
完了表示・完了欄へ移動
終了
設定画面
スプラッシュ／ログイン画面
ログイン／新規登録後
ホーム画面（タスク一覧）
設定ボタン
戻る
一覧更新
保存
キャンセル
＋ボタン
フェーズ詳細入力
タスク入力画面
日次フェーズ入力


TaskController
-taskService: TaskService
+create(request: Task): Task
+update(request: Task): Task
+delete(taskId: string): bool
+complete(taskId: string): bool
+search(query: string): List
TaskService
-taskRepository: TaskRepository
+createTask(task: Task): Task
+updateTask(task: Task): Task
+deleteTask(taskId: string): bool
+completeTask(taskId: string): bool
+searchTasks(query: string): List
NotificationController
-notificationService: NotificationService
+update(request: Notification): Notification
+send(userId: string): bool
TaskRepository
-tasks: List
+save(task: Task): Task
+findById(taskId: string): Task
+findByUserId(userId: string): List
+search(query: string): List
+delete(taskId: string): bool
User
+id: string
+email: string
+password: string
+createdAt: DateTime
+updatedAt: DateTime
+login(): bool
+logout(): void
+register(): bool
NotificationService
-notification: Notification
+updateNotification(notification: Notification): Notification
+sendNotification(userId: string): bool
Task
+id: string
+userId: string
+title: string
+goalDate: DateTime
+category: string
+status: TaskStatus
+createdAt: DateTime
+updatedAt: DateTime
+create(): Task
+update(): void
+delete(): void
+complete(): void
+search(query: string): List
Notification
+id: string
+userId: string
+isEnabled: boolean
+notificationTime: Time
+update(): void
+sendNotification(): bool
Phase
+id: string
+taskId: string
+phaseName: string
+startDate: DateTime
+endDate: DateTime
+order: number
+create(): Phase
+update(): void
+delete(): void
TaskStatus
+ACTIVE: string
+COMPLETED: string
+DELETED: string
+fromString(status: string): TaskStatus
+isTerminal(): bool
DailyTodo
+id: string
+phaseId: string
+content: string
+date: DateTime
+isCompleted: boolean
+create(): DailyTodo
+update(): void
+toggleComplete(): void
+delete(): void


USER
PK id String
UQ email String
NOT_NULL password String
NOT_NULL created_at Datetime
NOT_NULL updated_at Datetime
has
has
TASK
PK id String
FK user_id String
NOT_NULL title String
NOT_NULL goal_date Datetime
category String
NOT_NULL status String
NOT_NULL created_at Datetime
NOT_NULL updated_at Datetime
NOT_NULL start_date Datetime
NOT_NULL end_date Datetime
NOTIFICATION
PK id String
FK user_id String
NOT_NULL is_enabled Boolean
notification_time Time
NOT_NULL created_at Datetime
NOT_NULL updated_at Datetime
has
PHASE
PK id String
FK task_id String
NOT_NULL phase_name String
NOT_NULL start_date Datetime
NOT_NULL end_date Datetime
NOT_NULL order Number
NOT_NULL created_at Datetime
NOT_NULL updated_at Datetime
has
DAILYTODO
PK id String
FK phase_id String
NOT_NULL content String
NOT_NULL date Datetime
NOT_NULL is_completed Boolean
NOT_NULL created_at Datetime
NOT_NULL updated_at Datetime


User（ユーザー）
React UI（フロントエンド）
TaskController（APIコントローラー）
TaskService（ビジネスロジック）
TaskRepository（データアクセス層）
SQLite（データベース）
ホーム画面で＋ボタンをタップ
タスク入力画面を表示
タイトル・ゴール日・カテゴリを入力
フェーズ名・開始日・終了日を設定
各フェーズの日次ToDoを入力
保存ボタンをタップ
POST /api/tasks を送信
リクエストを受信
バリデーション実行
タスク保存
バリデーションエラー: 400 Bad Request
JSONエラーレスポンスを返却
エラー内容を表示
TaskをINSERT
DBエラー応答を生成
DB接続エラー: 500 Internal Server Error
Taskレコードを保存
保存結果を返却
PhaseレコードをINSERT（複数）
Phase保存結果を返却
DailyTodoレコードをINSERT（複数）
DailyTodo保存結果を返却
トランザクションをコミット
作成されたTaskオブジェクトを返却
コミット完了を返却
TaskServiceからTaskControllerへ返却
201 Created のJSONレスポンスを返却
ホーム画面へ遷移
新しいタスクを一覧に表示
User（ユーザー）
React UI（フロントエンド）
TaskController（APIコントローラー）
TaskService（ビジネスロジック）
TaskRepository（データアクセス層）
SQLite（データベース）

KASANE API Specification

Brief API for managing long-term goals via phases and daily tasks. RESTful JSON over HTTPS with JWT auth.

Overview

- Purpose: Manage goals → phases → daily tasks; track progress by date.
- Base URL: https://api.kasane.app
- Auth: Bearer JWT in Authorization header.

Authentication Endpoints

POST /api/auth/register

- Headers: Content-Type: application/json
- Body:

```
{"email":"user@example.com","password":"StrongPass123","name":"Aya"}
```

- 201:

```
{"user":{"id":"u1","email":"user@example.com","name":"Aya"},"token":"<jwt>"}
```

- 400/500: see Error Handling.

POST /api/auth/login

- Headers: Content-Type: application/json
- Body:

```
{"email":"user@example.com","password":"StrongPass123"}
```

- 200:

```
{"user":{"id":"u1","email":"user@example.com","name":"Aya"},"token":"<jwt>"}
```

- 401/400/500.

POST /api/auth/logout

- Headers: Authorization: Bearer
- 200: `{"message":"logged out"}`

Task Management Endpoints

POST /api/tasks

- Headers: Authorization, Content-Type: application/json
- Body:

```
{"title":"Thesis","goalDate":"2026-06-30","phases":[{"title":"Research","startDate":"2026-01-01","endDate":"2026-02-15","dailyTasks":[{"title":"Read 2 papers"}]}]}
```

- 201:

```
{"id":"t1",...}
```

- 400/401/500.

GET /api/tasks

- Headers: Authorization
- 200: `[{"id":"t1",...}]`

GET /api/tasks/:id

- Headers: Authorization
- 200: `{"id":"t1",...}`
- 404/401.

PATCH /api/tasks/:id

- Headers: Authorization, Content-Type
- Body (partial):

```
{"title":"Thesis Rev","phases":[...],"completed":false}
```

- 200: updated task
- 400/401/404.

DELETE /api/tasks/:id

- Headers: Authorization
- 204
- 401/404.

Daily Checklist Endpoints

PATCH /api/tasks/:id/check

- Headers: Authorization, Content-Type
- Body:

`{"phaseId":"p1","dailyTaskId":"d1","date":"2026-04-10","done":true}`

- 200: task with updated daily status
- 400/401/404.

GET /api/tasks/today

- Headers: Authorization
- 200: `[{"taskId":"t1","phaseId":"p1","dailyTaskId":"d1","title":"Read 2 papers"}]`

Search Endpoint

GET /api/tasks/search?q={keyword}

- Headers: Authorization
- 200: `[{"id":"t1","title":"Thesis"}]`
- 400/401.

Data Models

- User: `{id, email, name, passwordHash, createdAt, updatedAt}`
- Task: `{id, userId, title, goalDate, phases[], createdAt, updatedAt, completed}`
- Phase: `{id, title, startDate, endDate, dailyTasks[]}`
- DailyTask: `{id, title, occurrences:[{date, done}]}`

Status Codes Reference

- 200 OK, 201 Created, 204 No Content, 400 Bad Request, 401 Unauthorized, 404 Not Found, 500 Internal Server Error

Error Handling Format

`{"error":{"code":<int>,"message":"string","details":{}}}`

Notes

- Dates: ISO 8601 (UTC).
- JWT: short-lived access, refresh recommended.
- Rate limiting: 100 req/min/user; backoff on 429.
- CORS: allow https://app.kasane.app; Authorization, Content-Type headers.